

```
import streamlit as st
```

```
import pandas as pd
```

```
import os
```

```
FILE_PATH = "books.csv"
```

```
# Load or create CSV
```

```
def load_data():
```

```
    if os.path.exists(FILE_PATH):
```

```
        return pd.read_csv(FILE_PATH)
```

```
    else:
```

```
        df = pd.DataFrame(columns=["Book ID", "Title", "Author", "Status"])
```

```
        df.to_csv(FILE_PATH, index=False)
```

```
        return df
```

```
# Save into CSV
```

```
def save_data(df):
```

```
    df.to_csv(FILE_PATH, index=False)
```

```
st.title("Library Management System")
```

```
# Initialize session data
```

```
if "books" not in st.session_state:
```

```
    st.session_state.books = load_data()
```

```
menu = st.sidebar.selectbox("Select Option",
```

```
                             ["View Books", "Add Book", "Issue Book", "Return Book"])
```

```
# View all books
```

```
if menu == "View Books":
```

```
    st.header("Books List")
```

```
st.dataframe(st.session_state.books)
```

```
# Add new book
```

```
elif menu == "Add Book":
```

```
    st.header("Add Book")
```

```
    title = st.text_input("Book Title")
```

```
    author = st.text_input("Author Name")
```

```
    if st.button("Add"):
```

```
        if title and author:
```

```
            new_id = len(st.session_state.books) + 1
```

```
            new_book = pd.DataFrame({
```

```
                "Book ID": [new_id],
```

```
                "Title": [title],
```

```
                "Author": [author],
```

```
                "Status": ["Available"]
```

```
            })
```

```
            st.session_state.books = pd.concat([st.session_state.books, new_book], ignore_index=True)
```

```
            save_data(st.session_state.books)
```

```
            st.text("Book added successfully")
```

```
        else:
```

```
            st.text("Enter both Title and Author")
```

```
# Issue a book
```

```
elif menu == "Issue Book":
```

```
    st.header("Issue Book")
```

```
    available = st.session_state.books[st.session_state.books["Status"] == "Available"]
```

```
    if available.empty:
```

```
        st.text("No books available to issue")
```

```
    else:
```

```
book_id = st.selectbox("Select Book ID", available["Book ID"])

if st.button("Issue"):

    st.session_state.books.loc[st.session_state.books["Book ID"] == book_id, "Status"] = "Issued"

    save_data(st.session_state.books)

    st.text("Book issued successfully")
```

Return a book

```
elif menu == "Return Book":

    st.header("Return Book")

    issued = st.session_state.books[st.session_state.books["Status"] == "Issued"]

    if issued.empty:

        st.text("No issued books to return")

    else:

        book_id = st.selectbox("Select Book ID", issued["Book ID"])

        if st.button("Return"):

            st.session_state.books.loc[st.session_state.books["Book ID"] == book_id, "Status"] = "Available"

            save_data(st.session_state.books)

            st.text("Book returned successfully")
```